

AGNIBH DASGUPTA

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science	2021 – 2026
Utah State University, M.S., Computer Science	2018 – 2020
West Bengal University of Technology, B.Tech., Computer Science	2013 – 2017

Skills

Research Areas:	Representation Learning, Watermarking, Computer Vision, NLP, Multimodal Learning
Architectures:	Transformers, ViT, CLIP, LLMs, Diffusion Models
Programming:	Python, PyTorch, TensorFlow, Hugging Face Transformers
Distributed:	PyTorch DDP, DeepSpeed, FSDP, Slurm/HPC
Post-Training:	LoRA, SFT, DPO, GRPO, RLHF, RLAI
Agentic AI:	LangGraph, RAG, Tool Use, Function Calling
Evaluation:	LLM-as-a-Judge, RAGAS, A/B Testing, Human-in-the-Loop
Deployment:	vLLM, Quantization

Awards

CVPR Compute Transparency Champion Award	2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition	
• Awarded to TIACam for exemplary reporting of computational resources, methodology, and reproducibility details	

Professional Experience

University of Nebraska Omaha , Graduate Researcher	2021 – 2026
Representation learning, watermarking, and LLM research	
Utah State University , Graduate Researcher	2018 – 2020
Deep reinforcement learning research for financial trading applications.	

Selected Research and Projects

Siosa's Library: Path of Exile Q&A Agent	2026
• Multi-agent agentic workflow: LangGraph coordinates query expansion, tool selection across multiple data sources	
• Self-hosted a pretrained LLM on cloud infrastructure, optimized for inference using vLLM and quantization	
• Collected human preference pairs on retrieval quality and used them for post-training to align model behavior	
Invariant Features in LLMs: Geometric Characterization and Model Attribution (under review)	2026
• Identified invariant subspaces in 9 open-source LLMs via contrastive sensitivity decomposition	
• Validated semantic and nuisance components causally through next-token prediction interventions	
• Achieved 90%+ zero-shot model attribution accuracy across base, fine-tuned, and distilled variants	
TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation	CVPR 2026
• Built a zero-watermarking framework combining CLIP image-text features with a learned augmentation module	
• Trained a differentiable auto-augmentor to find camera-like distortions, removing hand-designed augmentation	
• Achieved 95%+ watermark recovery on ImageNet and Visual Genome under real-world camera conditions	
Watermarking Language Models through Language Models	IEEE TAI 2025
• Built a model-agnostic watermarking scheme for LLM-generated text without touching model weights	
• Tested on 25 open-source LLMs including fine-tuned and distilled variants using LoRA and DeepSpeed	
• Achieved 88%+ detection accuracy under paraphrasing, fine-tuning, and adversarial prompt attacks	
Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning	CSCI 2023
• Built a ViT-based cross-attention watermarking system with a self-supervised triplet loss objective	
• Trained the encoder to cluster augmented views while separating different images in latent space	
• Reached state-of-the-art comparable robustness under compound noise, confirmed via ablation	
Perspective Transformation Layer	CSCI 2022
• Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines	
• Evaluated on SVHN, Imagenette, and MNIST under viewpoint variation with minimal extra compute	
• Outperformed prior state-of-the-art on perspective benchmarks as a drop-in module	